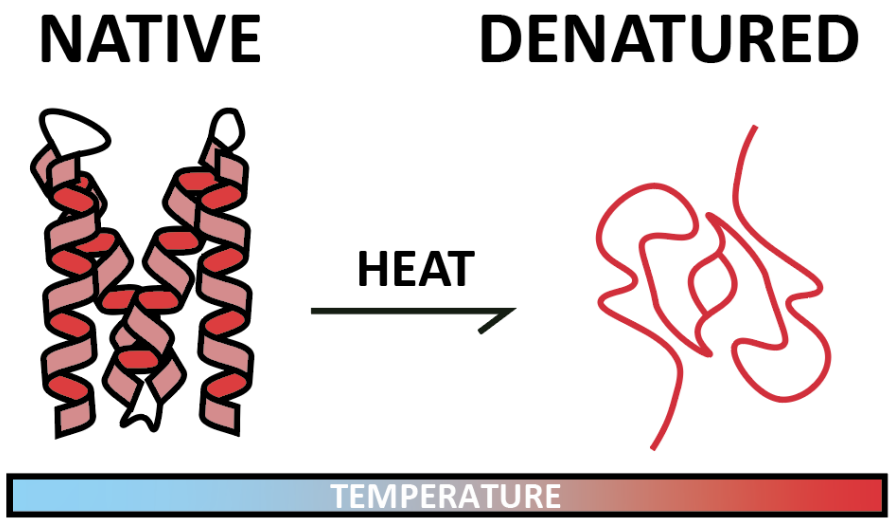
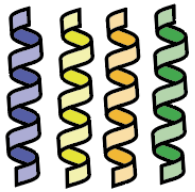


Tycholyzer

User Guide

By: Vishnu Ganesh Ghani

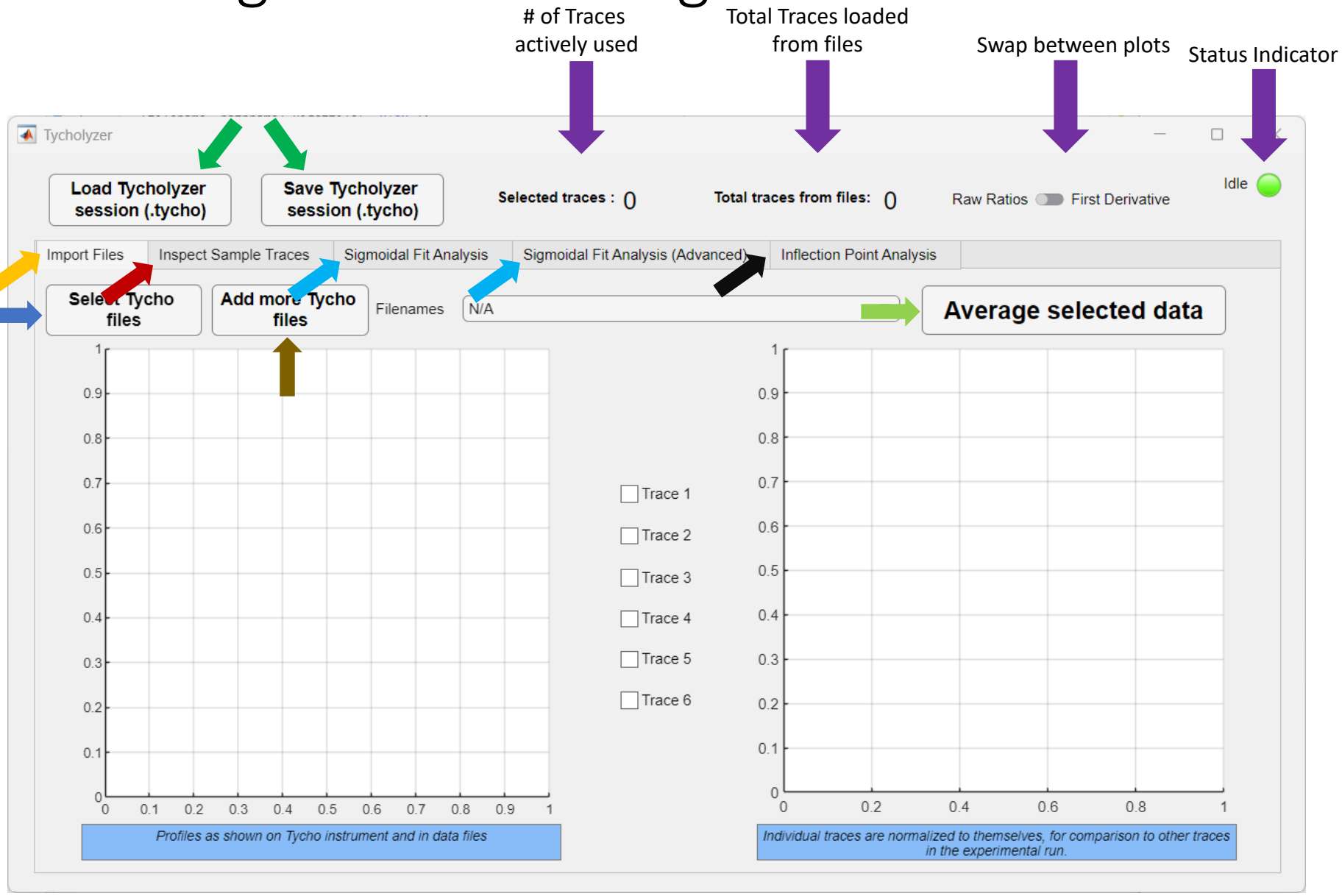
09/25/2026



Tycholyzer Background

- Standalone software that visualizes and analyzes NanoTemper Tycho data
  - Imports many files at once and shows normalized versions to combine and improve confidence of measurements
  - Tycho instrument only allows easy comparison of traces within the set measured in an experiment
- Sophisticated analysis for robust apparent melting temperature [T_M]
  - Curve fitting to single sigmoidal function or locates extrema from slopes of original ratios
  - Tycho only locates extrema from slopes of original ratios
- Do not need to install MATLAB to use
  - MATLAB Runtime (~2.7 GB) is built-in for the installation for software.
  - The Runtime is a one-time package to other MATLAB apps runs on

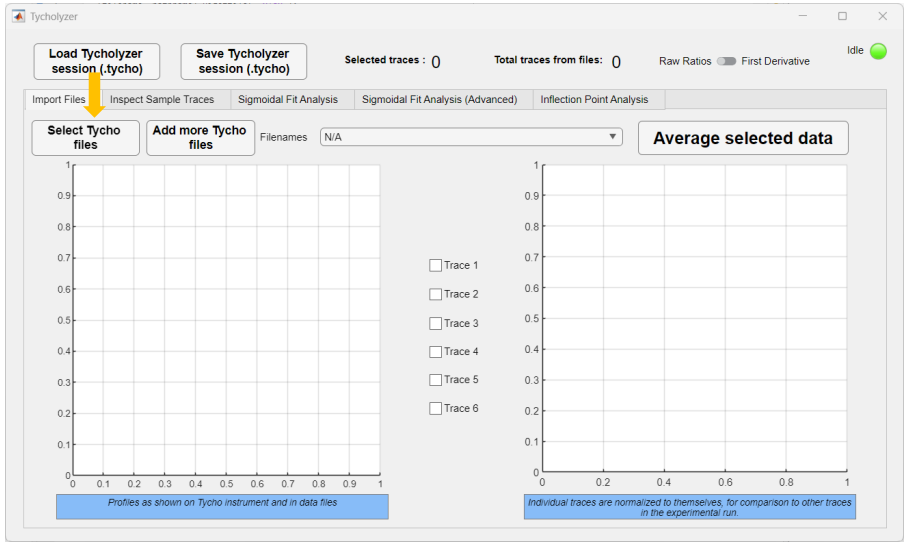
Screening Traces- Getting to know the Interface



- Load output tycho files the instrument
- Add more output tycho files to the currently loaded traces
- Customized session to load or save all your data in one place
- View and Filter traces from all files
- All traces that are still selected are grouped by their labels. Must press to have traces for 2nd , 3rd, and 4th tab
- Selected traces can qualitatively be viewed and compared
- Calculate T_Ms of each conditions and observe sigmoidal fit
- Get inflection points by determining local extrema

Screening Traces- Pipeline

1. Select tycho files



Load multiple files at once

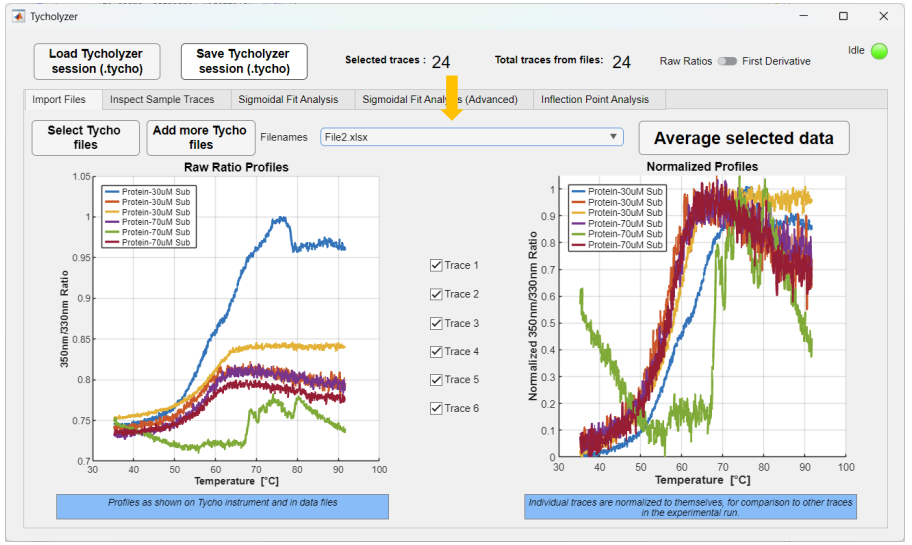
2. View and scan through traces



Left: Raw traces from files as shown on tycho

Right: Normalized-to-self versions of traces

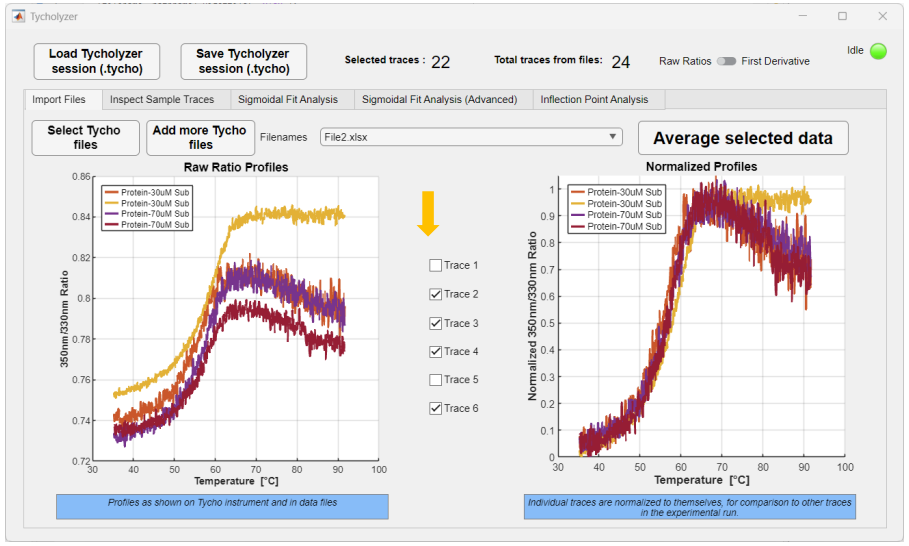
3. Select different files



Raster through the loaded files.

When a file is selected, the traces are linked to the corresponding checkbox if a capillary was observed in that position.

4. Unselect bad traces that shouldn't be in analysis

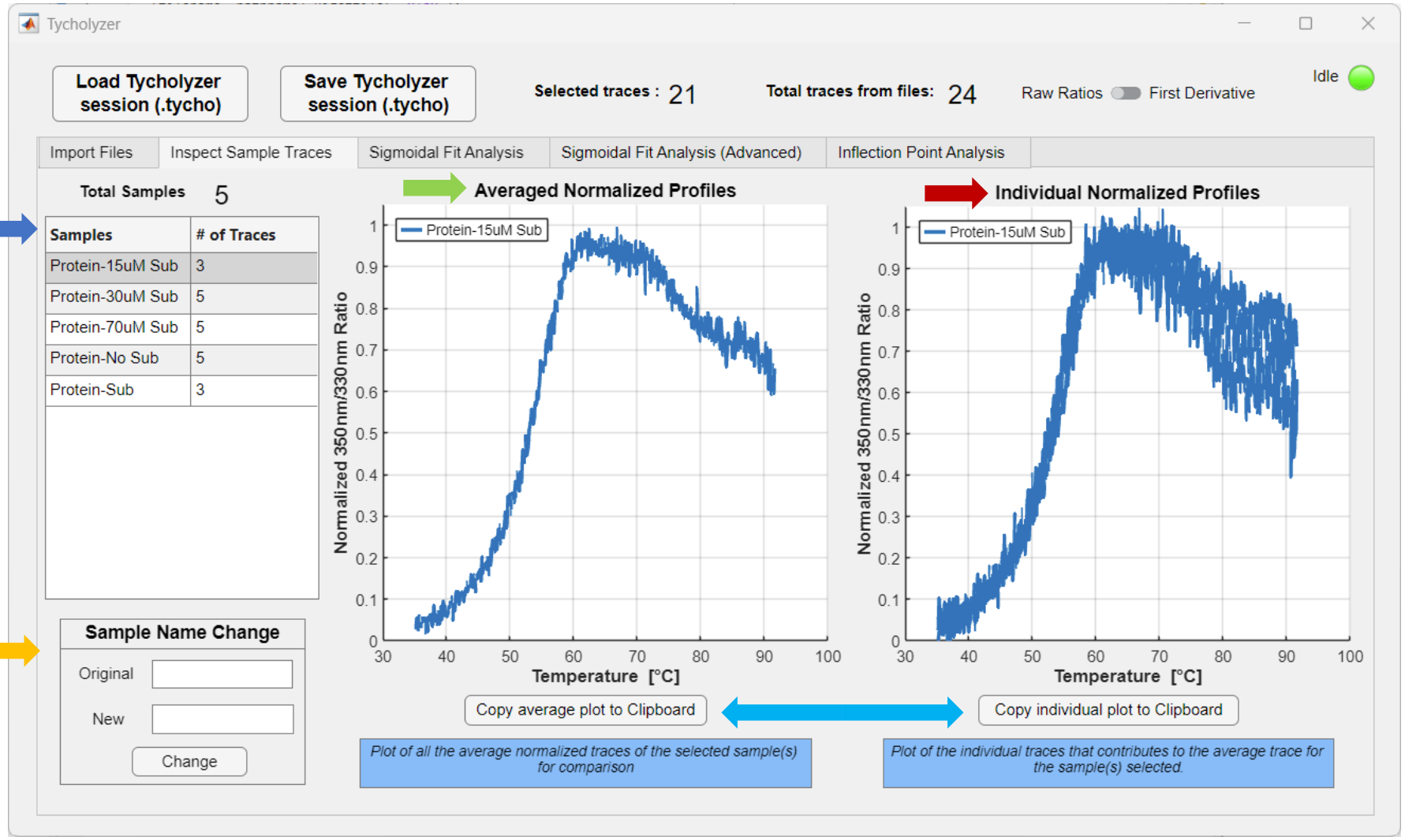


Unselect non-desirable traces.

Once you've decided that all remaining traces are appropriate, select [Average selected data].

These remaining traces are now the working data for the remaining tabs

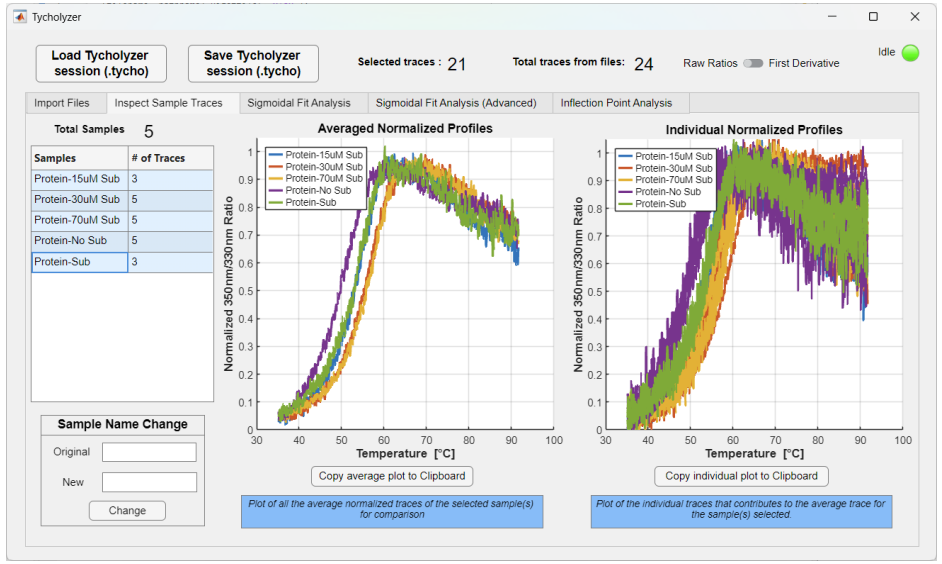
Condition Plots- Getting to know the Interface



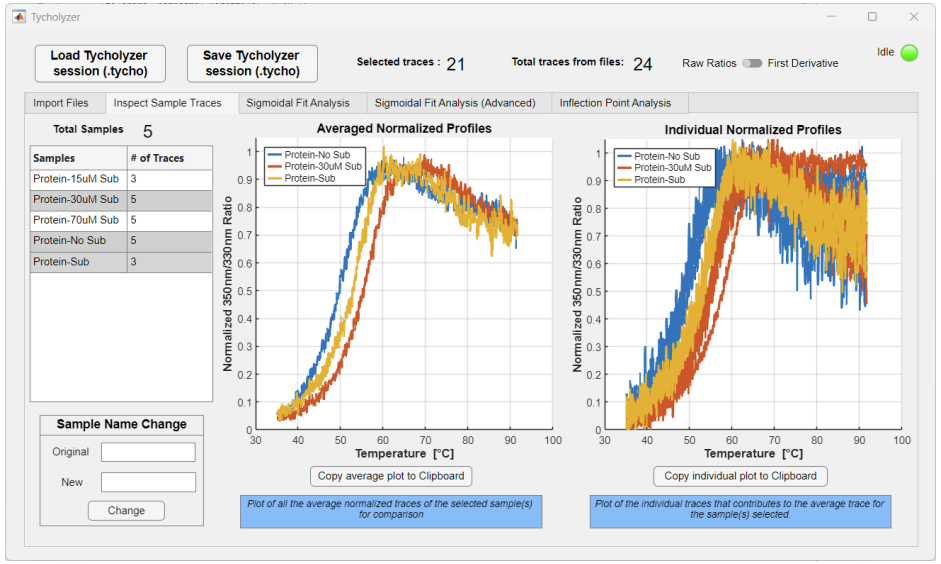
- Table of all the unique sample groups (trace names) and total number pooled together from all files.
- Ability to rename sample names. Works to combine samples as well.
- Shows the representative, average trace of the sample group
- Shows all the individual traces that goes into the average trace
- Copy the above graph for you to paste in any program

Condition Plots- Pipeline/Visualization

1. Multiple Plots Displayed



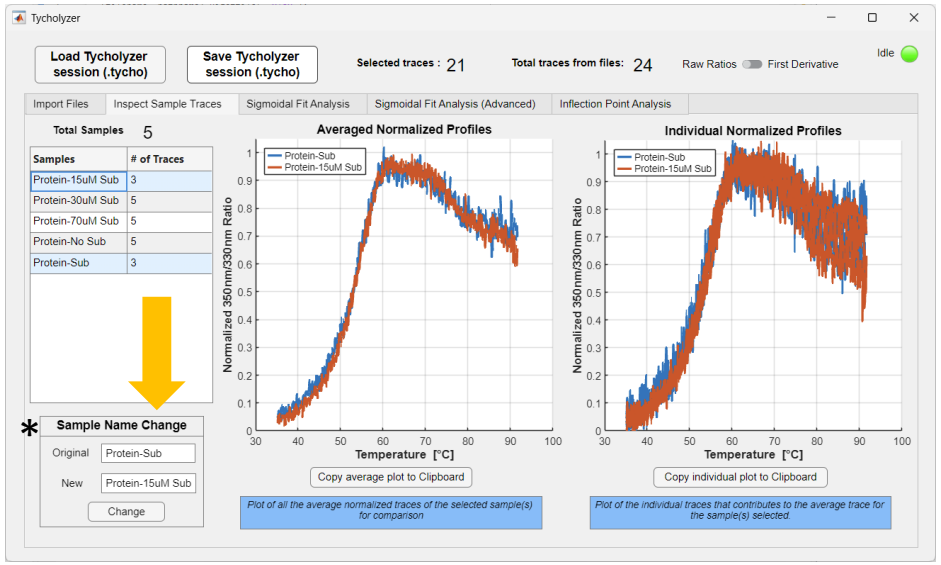
Multi-select many rows for display [maximum of 12 conditions can be plotted at the same time]



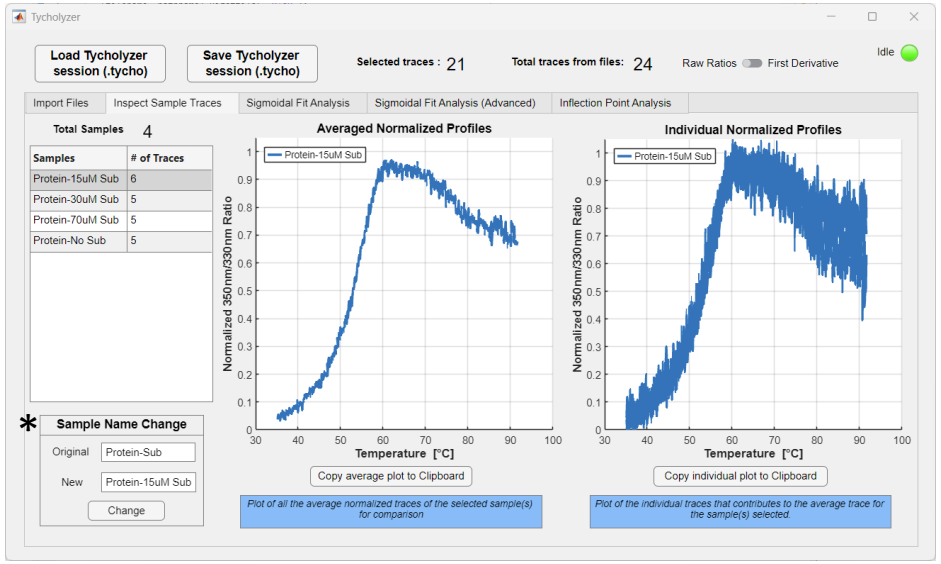
Selections are order-sensitive if you need to preserve color scheme for traces. Can change order of selection left clicking and holding ctrl key

e.g.: Protein-No Sub, Protein-30uM Sub, and Protein-Sub were selected in that order

2. Sample Name Change



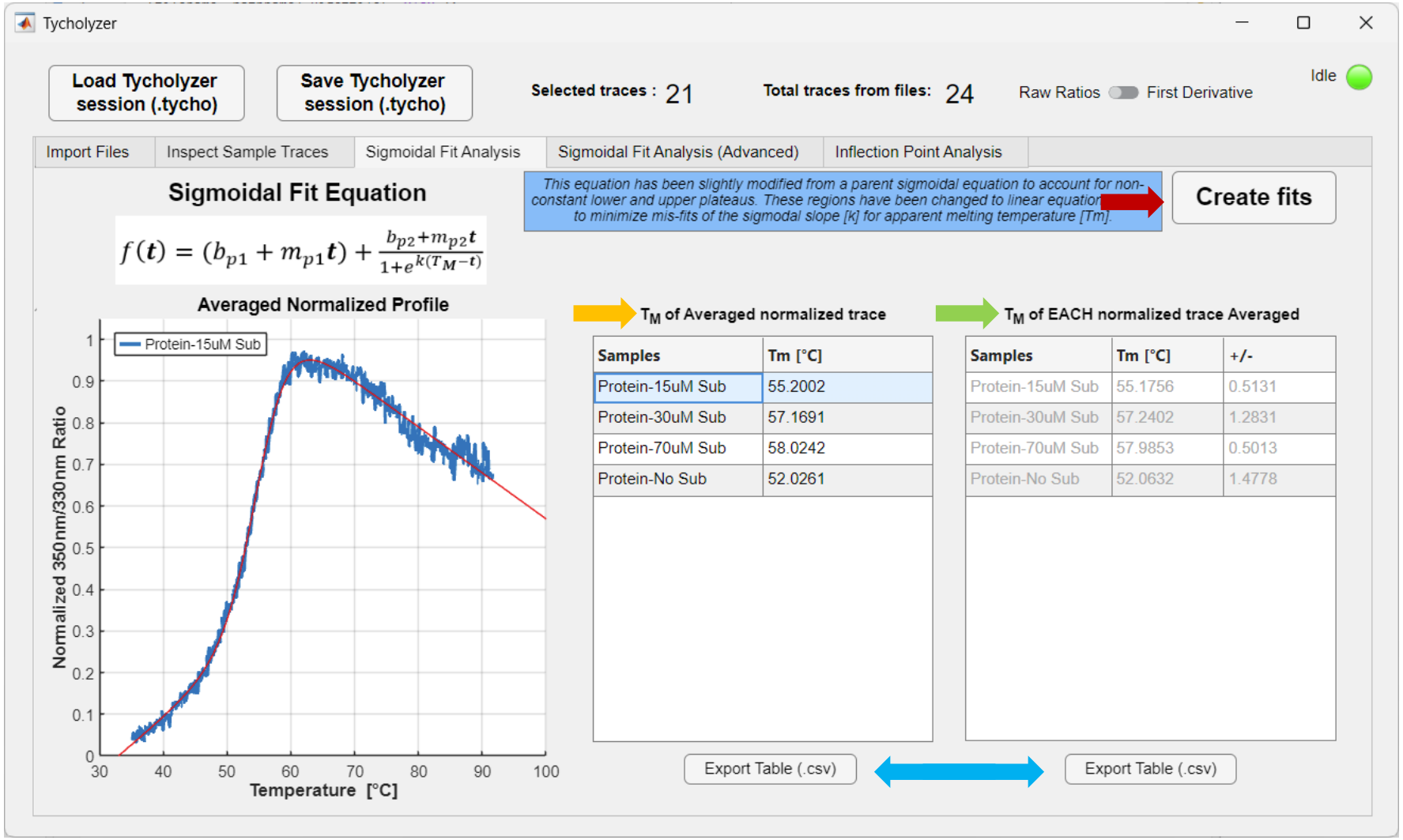
In this tutorial, Protein-15uM Sub was accidentally was named as Protein-Sub in one of the files. Instead of trying to manually changing all the names in all the raw files, we can change it this tab



We can now see that the number of conditions went down by 1, and the # of traces for Protein-15uM Sub went from 3 to 6

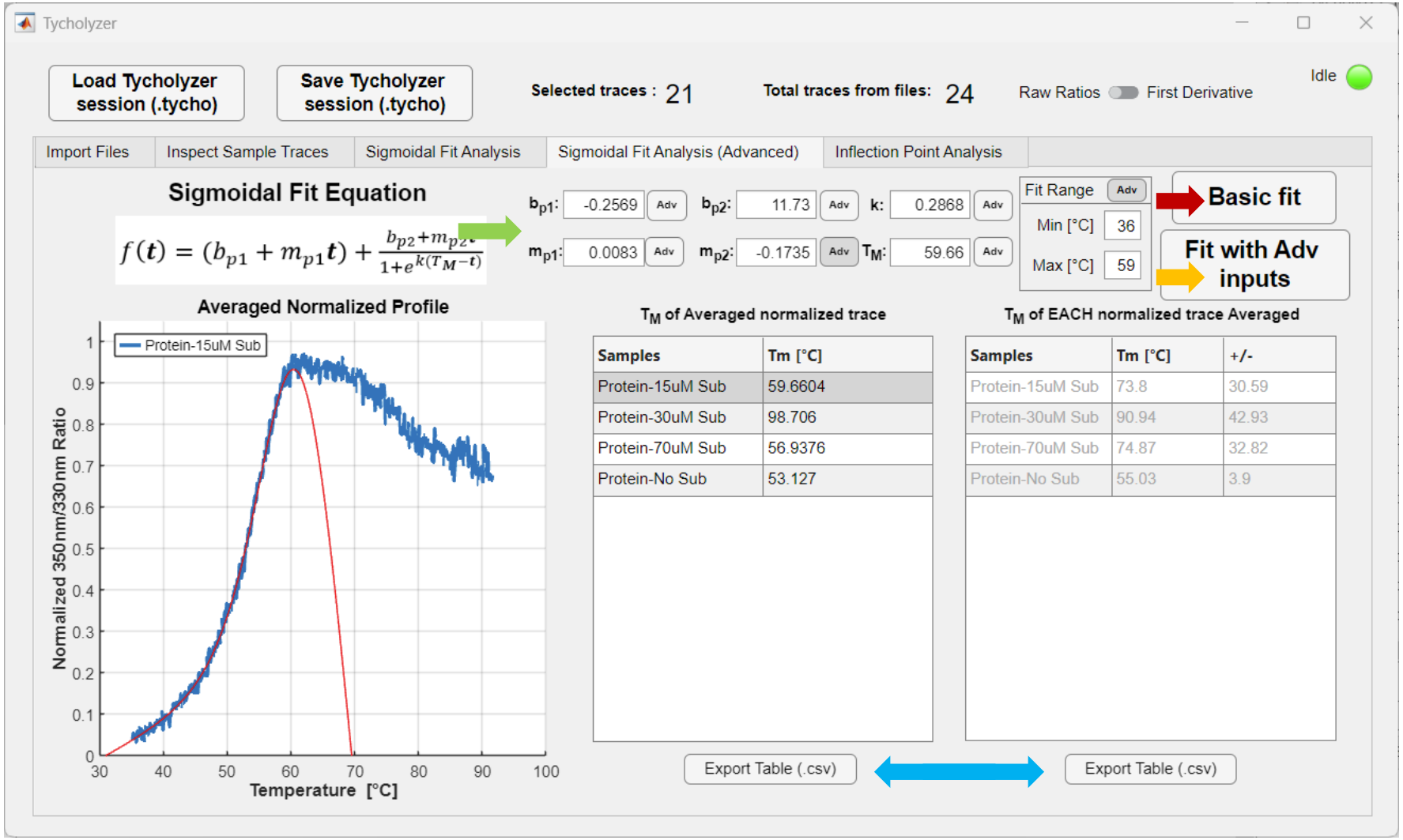
*You can copy and paste the selected row in the table into the Condition Name Change boxes

Fit Analysis- Getting to know Interface



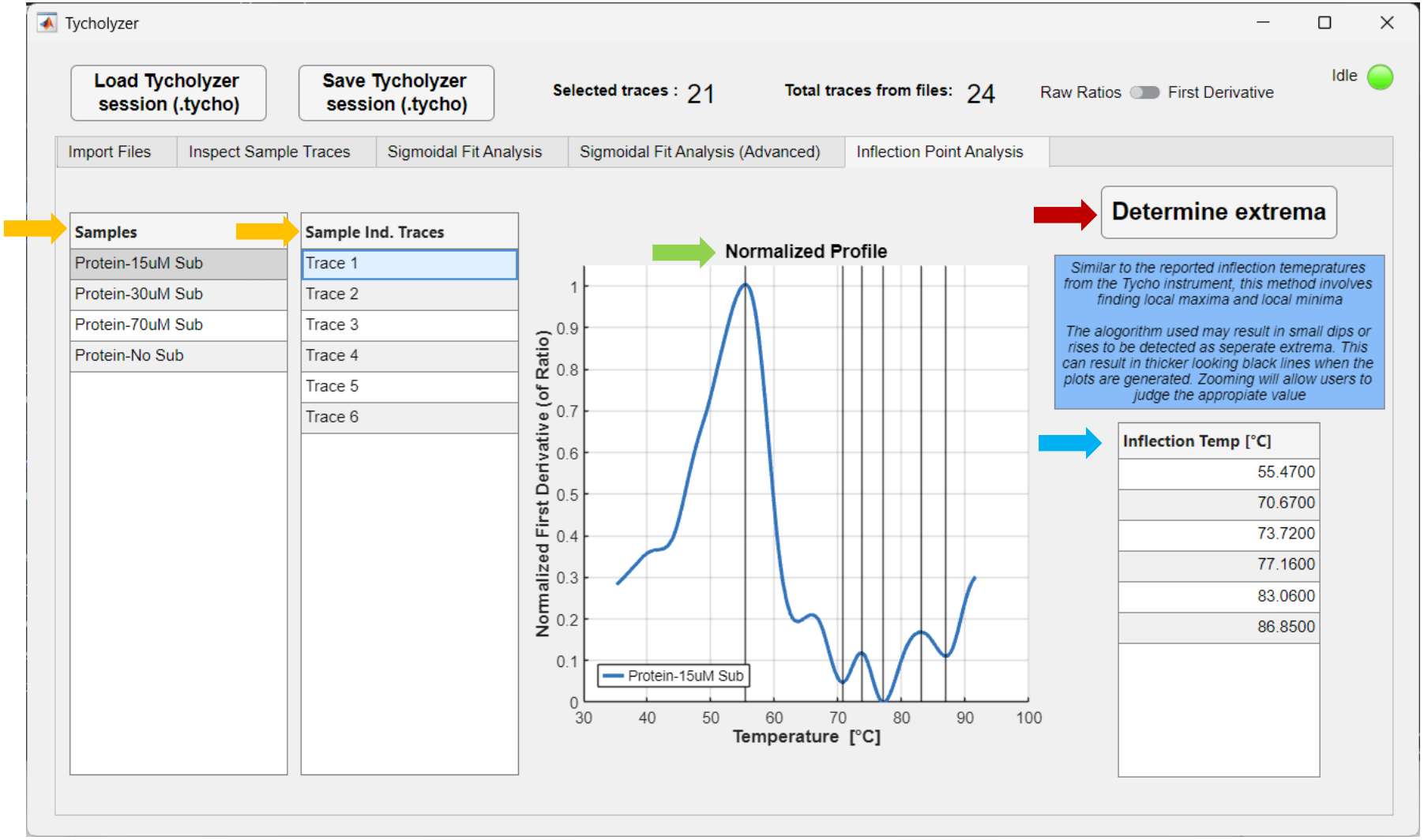
- Select to generate fits
- Table of calculated T_M from the average trace of each condition. Selecting a row will display the average trace and the fitted curve (red line). Only one will be displayed at a time
- Table of calculated T_M by fitting to each individual trace of the condition, then averaging the values. +/- is standard deivation
- Export table values into a csv file

Fit Analysis (Advanced)- Getting to know Interface



- Select to fit like in previous tab
- Perform fit on selected trace with whichever Adv parameters pushed down for initial guesses around inputted values
- Parameters for users to edit to improve sigmodal fits
- Export table values into a csv file

Inflection Identification- Getting to know Interface



- Select to generate extrema calculation
- Tables of different samples groups, and individual traces
- Graph of selected condition 1st derivative with vertical lines denoting local extrema detected
- Table of corresponding temperatures for the vertical lines denoting local extrema